



Read the files:

1. "Final Project [Guidelines.md](#)"
2. "Final Project proposal - Steering Idioms [Extension.md](#)", which references the "Steering Idioms- Controlling Figurative-vs-Literal Interpretation via Residual-Stream Activation Steering.pdf" article. Side note - IdioLink can be found here <https://arxiv.org/abs/2605.22247>

Our goal here is to plan the execution of our project. We need do-able step-by-step plan. We prefer not to be blocked by talking to the instructor Kai if possible (if we have any dependency on him - lets draft relevant message to him, bu also lets try to have fallback for us to be unblocked).

Ignore previous conversations for answering this one.

You can execute this project in 6–7 focused steps over the next ~2 weeks, even without ongoing input from Kai, by first reproducing idiom steering, then extending it to a small figurative (non-idiom) benchmark and wrapping everything into the required report, slides, and video.

Project recap and constraints

The approved proposal is to test whether the idiom-based steering direction from *Steering Idioms* (IdioSteer) reflects a general figurative–literal axis that also applies to non-idiomatic figurative expressions (metaphors, similes etc.), rather than being idiom-specific.^{[1] [2]}

The course requires an 8-page-max research-style report, plus slides and a 5-minute presentation video, all due 20/8/2026, with strong weight on clear problem definition, methodology, novelty, and analysis.^[3]

IdioLink is a separate idiom-aware retrieval benchmark focusing on linking idiomatic spans to conceptually equivalent literal/paraphrased meanings, with 10,700 documents and 2,140 queries over 107 idioms.^{[4] [5]}

Step 0 – Fix scope, assumptions, and timeline

1. Scope decision (today)
 - Core deliverable: monolingual English steering on idioms + non-idiomatic figurative benchmark (metaphor/simile), on **one base LLM** (optionally a second for robustness).
 - Optional extensions only if time: a second figurative category (e.g. metaphor vs simile) or a second model family.
2. Compute and model assumptions

- Assume access to a single GPU (local or Colab) that can comfortably run a 7–9B open-weights LLM (e.g., LLaMA or Gemma), which matches the Steering Idioms setup using mid-size models from one or two families.^[2]
- Plan steering implementation directly in PyTorch/Transformers: intercept residual stream, add steering vector at chosen layers and timesteps, following standard activation-steering practices.^{[6] [7]}

3. Timeline sketch (relative to today)

- Days 1–2: Baseline reproduction of idiom steering (IdioSteer-style).
- Days 3–5: Design and annotate non-idiomatic figurative benchmark; generate steered/unsteered continuations.
- Days 6–8: Analysis, optional new steering direction, geometry comparison.
- Days 9–11: Write report.
- Days 12–13: Slides + 5-minute video + polishing and contingency runs.

Step 1 – Reproduce idiom steering baseline (IdioSteer)

Goal: Get a minimal working pipeline that recovers the core Steering Idioms result: steering a residual-stream vector pushes continuations from figurative toward literal readings on idioms, while preserving fluency.^[2]

1. Paper re-read and protocol extraction (few hours)

- Carefully re-read *Steering Idioms* to extract:
 - How idiom stimuli are constructed (100 sentences, 20 idioms, figurative vs literal contexts).
 - How residual activations are collected (which layer, which token positions, which prompt format).
 - How steering vectors are computed (difference between figurative vs literal activations, averaged across examples).
 - Where in the forward pass the steering is injected (specific layers, timesteps) and coefficient grid used.^{[7] [2]}

2. Dataset: practical approach

- If IdioSteer stimuli are publicly released (check any linked repo or data URL in the paper): download and use directly.
- If not public or hard to access: manually reconstruct a **small subset**, e.g. 10–12 idioms × 4 contexts each (figurative/literal, varied prefixes).
 - Use idioms listed or exemplified in the paper (e.g., “break the ice”, “bad apple”) and follow their contextualization style.^[2]
 - Keep this set compact enough to handle within the project window.

3. Model and code setup (1 day)

- Choose one open LLM (e.g., LLaMA-family base model) with accessible residual stream hooks.
- Implement a small wrapper that:
 - Runs generation with a fixed decoding setup (same for all experiments).
 - Records residual activations for chosen layer(s) and positions.
 - Allows adding an arbitrary steering vector v scaled by coefficient α at those layers during generation.

4. Construct idiom steering vector

- For each idiom stimulus pair (figurative vs literal context), compute the residual difference $d_i = r_i^{\text{literal}} - r_i^{\text{figurative}}$ at the chosen layer/token. ^[2]
- Average across examples to get a single idiom steering vector $v = \frac{1}{N} \sum_i d_i$.
- Probe cross-layer injection (vector from mid/late layer applied at earlier layer) vs same-layer injection, since cross-layer steering was found to perform better in the original paper. ^[2]

5. Baseline evaluation on idioms

- For each idiom prefix:
 - Generate unsteered continuation.
 - Generate steered continuation at several coefficients (e.g., $\alpha \in \{-2, -1, 0, +1, +2\}$).
- Label figurative vs literal reading with an LLM-based classifier plus a small human-checked subset, mirroring Steering Idioms evaluation. ^{[1] [2]}
- Confirm qualitative effect: steering increases literal readings without breaking fluency, and transfers to held-out idioms. ^[2]

If this phase works, you already have a solid replication and a usable steering vector; if anything breaks, you still have the code and dataset needed for the extension.

Step 2 – Build non-idiomatic figurative benchmark

Goal: A compact, hand-crafted benchmark of non-idiomatic figurative expressions (metaphors/similes) where both figurative and literal continuations are plausible, with unambiguous labeling of target reading.

1. Define categories and size (half day)

- Focus on **metaphors and similes** (simple enough, close to idioms but not compositional multiword expressions).
- Target size: ~40–60 items (e.g., 30 metaphors + 20 similes), enough for analysis but still manageable.

2. Design stimulus format

- For each expression, write:

- A prefix that sets up ambiguity (either figurative or literal reading is plausible).
- A “figurative” gold continuation and a “literal” gold continuation, used for labeling templates and sanity checks.
- Aim for parallel structure to IdioSteer: prefixes ending at the figurative expression, followed by continuation generated by the model. ^[1] ^[2]

3. Annotation procedure

- For each item, annotate:
 - Expression type (metaphor vs simile).
 - Intended figurative concept (e.g., “time is a thief” → concept: “time takes away valued things”).
 - Literal reading description.
- Use these annotations to guide later LLM-based labeling and human sanity checks.

4. Optional IdioLink tie-in

- As a light extension, you can borrow idioms and literal paraphrases from IdioLink to ensure your new benchmark’s figurative/literal contrasts align with concept-level distinctions already used in retrieval. ^[5] ^[4]
- This is optional; the core project doesn’t depend on integrating retrieval, but it strengthens the conceptual link in the report.

Step 3 – Apply idiom steering to the new figurative benchmark

Goal: Test whether the idiom-derived steering direction alters figurative vs literal rates on non-idiomatic figurative expressions in a systematic way.

1. Generation setup

- Use the same model, prompting style, and decoding parameters as in the idiom experiments to keep comparisons clean. ^[1] ^[2]
- For each figurative prefix in your new benchmark:
 - Generate unsteered continuation.
 - Generate steered continuations at the same coefficient grid used for idioms.

2. Labeling figurative vs literal

- Adapt the LLM-based labeling protocol from Steering Idioms:
 - Provide the prefix and continuation to a labeling model with a clear instruction to classify whether the continuation interprets the expression literally or figuratively. ^[2]
 - Calibrate prompts with a few examples from your gold figurative/literal continuations.
- Manually inspect a subset (e.g., 10–15 items) to verify labeling quality and adjust the prompt if needed.

3. Metrics

- Compute, for idioms and for your new benchmark:
 - Figurative rate and literal rate under each steering coefficient (Target Category Rate in their terminology).
 - Change in literal rate between unsteered and steered runs.
- Check whether the **same steering vector**:
 - Increases literal readings on idioms (as in the original paper).^[2]
 - Has a consistent effect (increase or decrease) on metaphors/similes.

4. Qualitative analysis

- Collect representative examples where steering:
 - Cleanly flips a metaphor to a more literal description.
 - Fails or produces odd behavior (e.g., nonsensical literalization or mixing figurative and literal aspects).
- These examples will be useful for the “Results and Analysis” and “Methodology” sections in the final report.^[3]

Step 4 – Optional: learn a new figurative steering direction and compare geometry

If time allows, add a small but high-impact extension: derive a steering vector from the non-idiomatic benchmark itself and compare it to the idiom-based vector.

1. Construct figurative–literal difference vector from new data

- For each metaphor/simile item, collect residual activations for contexts that bias toward figurative vs literal readings (e.g., by constructing two prompts or using your gold continuations).
- Compute differences and average as before to get a “figurative steering” or “literalizing” vector specific to non-idiomatic expressions.

2. Behavior comparison

- Apply:
 - Idiom-based vector to idioms and metaphors/similes.
 - Non-idiom vector to idioms and metaphors/similes.
- Compare changes in figurative/literal rates: which vector generalizes better across expression types?

3. Geometry comparison

- Compute cosine similarity between the idiom-based vector and the non-idiom vector; check whether they are aligned, orthogonal, or somewhere in between.
- If possible, visualize both vectors in a low-dimensional projection or analyze which neurons/layers contribute most, connecting to broader work on activation steering and steerability manifolds.^{[6] [7]}

Even a rough comparison here supports the “Novelty” and “Interpretation” criteria without requiring large-scale experiments.^[3] ^[1]

Step 5 – Analysis and writing the report

Goal: convert the above pipeline and results into a clear 6–8-page report matching the grading criteria and Overleaf template.^[3]

1. Report structure (aligned with template)

- Introduction: idioms, figurative vs literal ambiguity, activation steering, Steering Idioms, and the generalization question.
- Background & related work: Steering Idioms, IdioLink (retrieval side), and relevant activation-steering papers.^[4] ^[7] ^[6] ^[2]
- Methodology:
 - Models and implementation details (layers, hooks, coefficient grid, labeling protocol).
 - Idiom dataset (IdioSteer replication or subset).
 - New figurative benchmark design, examples, and annotation.
- Experiments:
 - Baseline idiom steering results.
 - Steering behavior on metaphors/similes.
 - Optional new steering vector and geometry.
- Results & analysis: quantitative metrics, qualitative examples, limitations.
- Conclusion: does the idiom steering vector behave like a general figurative-literal axis?

2. Grading alignment

- Explicitly highlight:
 - Problem definition and motivation (figurative vs literal control beyond idioms).
 - Novelty: new benchmark + cross-category steering probe.
 - Methodology: detailed steering protocol, including a diagram of the residual-stream intervention.
 - Results and analysis: including both successes and failure cases.
- Keep the report within 8 pages, as specified.^[3]

Step 6 – Slides and 5-minute video

Goal: simple, concise communication of the project for grading and future re-use.

1. Slides (1 day)

- 8–12 slides:

- Motivation & problem.
- Steering Idioms recap.
- Your pipeline diagram.
- Idiom baseline results.
- New figurative benchmark.
- Steering behavior across idioms vs metaphors/similes.
- Optional geometry comparison.
- Takeaways and limitations.

2. Video (half day)

- Record a 4–5 minute walkthrough of the slides, focusing on intuition rather than all details.
- Check that the video and slides together are self-explanatory, as required. ^[3]

Risk management and fallback paths

You can largely avoid being blocked by Kai by designing the plan to be self-contained and by having explicit fallbacks.

1. If IdioSteer data or code is not accessible

- Reconstruct a smaller idiom dataset yourself (10–12 idioms) and follow the activation-steering method described in the paper, rather than relying on official code. ^[2]
- This slightly reduces scale but preserves the core experimental logic.

2. If steering implementation proves tricky

- Fall back to a simpler protocol: record residual differences and apply vectors only at one layer and one timestep; focus on idiom vs non-idiom qualitative examples rather than exhaustive coefficient grids.
- Emphasize limitations and connect to theoretical work on steerability and non-surjectivity to show awareness of technical constraints. ^[7]

3. If time runs short

- Ensure at least:
 - Idiom steering replication on a small set.
 - Metaphor/simile benchmark with at least one steering condition.
 - Clear narrative and analysis of what works and what appears idiom-specific.

This still yields a coherent research project clearly aligned with the course guidelines. ^{[1] [3]}

Optional email draft to Kai (for clarification, but not a hard dependency)

If you decide you need one round of clarification or approval, you can send something like this; but the plan above is designed so you can proceed even without a reply.

Subject: Final project – Steering Idioms extension (clarifications & small benchmark)

Hi Kai,

Following your endorsement of my project proposal on extending *Steering Idioms* beyond idioms, I wanted to briefly confirm that my planned scope is aligned with course expectations and doesn't rely on heavy resources.

Concretely, I plan to:

- Reproduce a minimal version of the idiom steering setup from the paper (on one open-weights LLM and a compact subset of idioms).
- Build a small benchmark of non-idiomatic figurative expressions (mostly metaphors and similes) where both figurative and literal continuations are plausible.
- Apply the idiom-derived steering vector to this new benchmark and compare figurative vs literal rates, plus some qualitative examples.
- If time allows, learn a second steering direction from the non-idiomatic data and compare its geometry and behavior to the idiom one.

I'll keep the experiments lightweight (single family of base models, compact datasets) and focus on clear analysis and limitations rather than scale. If there is any particular aspect of the Steering Idioms protocol you think I should definitely include or avoid (e.g., specific layers, evaluation details, or dataset usage), I'd really appreciate a quick note.

Otherwise, I'll proceed with this plan and aim to tie the results back to your recent idiom work and, where relevant, IdioLink.

Thanks in advance,

[Your Name]

You can send this once, but continue executing the steps above regardless of response, so you're never blocked.

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1. [Final-Project-proposal-Steering-Idioms-Extension-2.md](#)
2. [Steering-Idioms-Controlling-Figurative-vs-Literal-Interpretation-via-Residual-Stream-Activation-.pdf](#)
3. [Copy-of-Final-Project-Guidelines-3.md](#)
4. <https://arxiv.org/abs/2605.22247>
5. <https://arxiv.org/html/2605.22247>
6. <https://www.themoonlight.io/en/review/improving-instruction-following-in-language-models-through-activation-steering>
7. <https://iclr.cc/virtual/2026/10016066>
8. <https://www.semanticscholar.org/paper/IdioLink:-Retrieving-Meaning-Beyond-Words-Across-Hashiloni-Fadlon/ef30b8bfb906eb3f7e8a1afed3a1481666710c31>
9. <https://cdn-avatars.qwak.ai/datasets/Intellexus/IdioLink>

10. <https://liner.com/ko/review/improving-instructionfollowing-in-language-models-through-activation-steering>
11. <https://openreview.net/pdf?id=E8XhJDFcjZ>
12. <https://aclanthology.org/2026.mwe-1.12.pdf>
13. <https://arxiv.org/html/2605.30443v1>
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